Together with iGEM

Across Villages, From Myths to Facts.

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Climate Crisis

Conservation

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Preface

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When learning about synthetic biology, we often encounter claims that sound plausible but are not entirely accurate. To help readers understand these misconceptions, JLU-FBH launched Together with iGEM, inviting each participating iGEM team to identify common misconceptions within its chosen village and provide clear explanations.

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This handbook brings together their contributions, collected, organized, and laid out by JLU-FBH. Each entry presents a misconception alongside the facts, explaining why the claim is misleading and how to understand the topic more accurately.

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The teams cover a range of fields, from agriculture and food to health, the environment, art, computing, and space. Their contributions offer readers an introduction to different villages through specific questions. Whether you are new to synthetic biology or already familiar with it, we hope you find something that interests you or gives you a new perspective.

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Together with iGEM reflects the shared effort behind this handbook. We thank every participating team for the time and care they put into their contributions. We hope these explanations reach more readers, answer questions, and encourage a closer look at ideas we often take for granted.

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Editor-in-Chief Team: JLU-FBH

In collaboration with 17 iGEM teams worldwide

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The co-editor profiles

(Listed in no particular order.)

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All tumors are cancer.

The Truth:

Medically, a tumor is a mass formed by abnormal cell proliferation, classified into three major types: benign, borderline , and malignant. Among these, malignant tumors are what the public commonly refers to as "cancer"—which includes carcinomas, sarcomas, and hematologic malignancies such as leukemia. Malignant tumors invade surrounding tissues and metastasize, posing a life-threatening risk. In contrast, benign tumors typically grow slowly, do not metastasize, do not invade surrounding tissues, and rarely recur after surgical excision—fundamentally distinct from "cancer." Conflating the two can either cause unnecessary panic or lead to delayed treatment.

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The smaller the tumor, the safer it is; tumor risk is directly proportional to its size.

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The Truth:

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The danger of a tumor primarily depends on its malignancy grade and molecular subtype, not simply its size. Some small triple-negative or HER2-positive breast cancers can be highly aggressive and metastasize early, while some larger ductal carcinomas in situ or indolent subtypes may grow slowly with favorable prognosis. In synthetic biology-targeted therapies, treatment strategies are often determined by molecular biomarkers rather than tumor size.

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After surgical removal of the tumor, the patient is completely cured and no longer needs to worry about recurrence.

The Truth:

Surgery removes the visible solid tumor mass, but patients may still have minimal residual disease (MRD) or circulating tumor cells (CTCs) undetectable by imaging, which are potential sources of long-term recurrence. Therefore, postoperative systemic therapies—including chemotherapy, radiotherapy, targeted therapy, endocrine therapy, or immunotherapy—are often needed based on pathological and molecular subtyping to eliminate residual cancer cells, along with long-term regular follow-up monitoring.

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Sex differences in cancer are only about reproductive system tumors—immune differences between sexes are negligible.

The Truth:

Sex is a severely underestimated variable in tumor immunotherapy. Women generally have stronger innate and adaptive immune responses, which is advantageous against infections but also results in higher autoimmune disease rates. In tumor immunotherapy, female patients show significantly different responses and adverse reaction profiles to immune checkpoint inhibitors compared to males. This difference is closely linked to the dosage effect of immune-related genes on the X chromosome and sex hormone regulation of immune cells—especially important in breast cancer treatment.

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Sugar feeds cancer cells, so cancer patients should completely eliminate all sugar from their diet.

The Truth:

All human cells—including normal cells—require glucose for energy. Complete sugar elimination leads to severe malnutrition and weight loss, accelerating cachexia. The correct approach is to control excessive intake of refined and added sugars (which benefits overall health), not total sugar elimination. Nutritional support is a critical component of cancer care—blindly cutting sugar does more harm than good.

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Alkaline foods can kill cancer cells, and acidic body pH is the root cause of cancer.

The Truth:

"acid-alkaline body" theory has no scientific basis and has been officially debunked by multiple scientific institutions. Human blood pH is tightly regulated within 7.35–7.45 and cannot be significantly altered by dietary intake—the kidneys and respiratory system maintain powerful acid-base homeostasis. The so-called "acidity/alkalinity" of foods cannot change internal body pH, let alone kill cancer cells. Blindly pursuing "alkaline diets" may cause nutritional imbalances and delay proper treatment.

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Microwaving food produces carcinogens and increases cancer risk.

The Truth:

Microwave ovens use non-ionizing radiation to generate heat through water molecule vibration—this thermal effect is fundamentally no different from conventional heating and does not induce radiological or mutagenic changes in food molecules, nor generate carcinogens. Microwave heating is comparable to steaming and in some cases better retains water-soluble vitamins. The electromagnetic frequency of microwave ovens is far below the energy threshold for ionizing radiation. No credible scientific evidence shows normal microwave use increases cancer risk.

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Natural foods like lemon juice, baking soda, or ginger can kill cancer cells within hours—safer and more effective than chemotherapy.

The Truth:

The "natural miracle cure" myth resurfaces every year repackaged with different ingredients and is highly deceptive. In vitro "cancer cell killing" mostly involves applying high-concentration extracts directly to cancer cells in petri dishes—completely different from the complex in vivo processes of digestion, absorption, metabolism, and accumulation to therapeutic concentrations. Salt solution can also kill cells in the lab, but no one treats cancer with saltwater. Exaggerating the anticancer efficacy of natural foods and inducing patients to forgo or postpone proper treatment constitutes a serious threat to patient safety.

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Breast cancer only occurs in women.

The Truth:

Although breast cancer is far more common in women, men also have breast tissue and can develop breast cancer. Male breast cancer accounts for approximately 1% of all breast cancer cases. However, due to lack of awareness, male patients are often diagnosed at later stages with worse prognosis. Anyone noticing abnormal breast lumps or nipple discharge should seek medical attention promptly regardless of gender.

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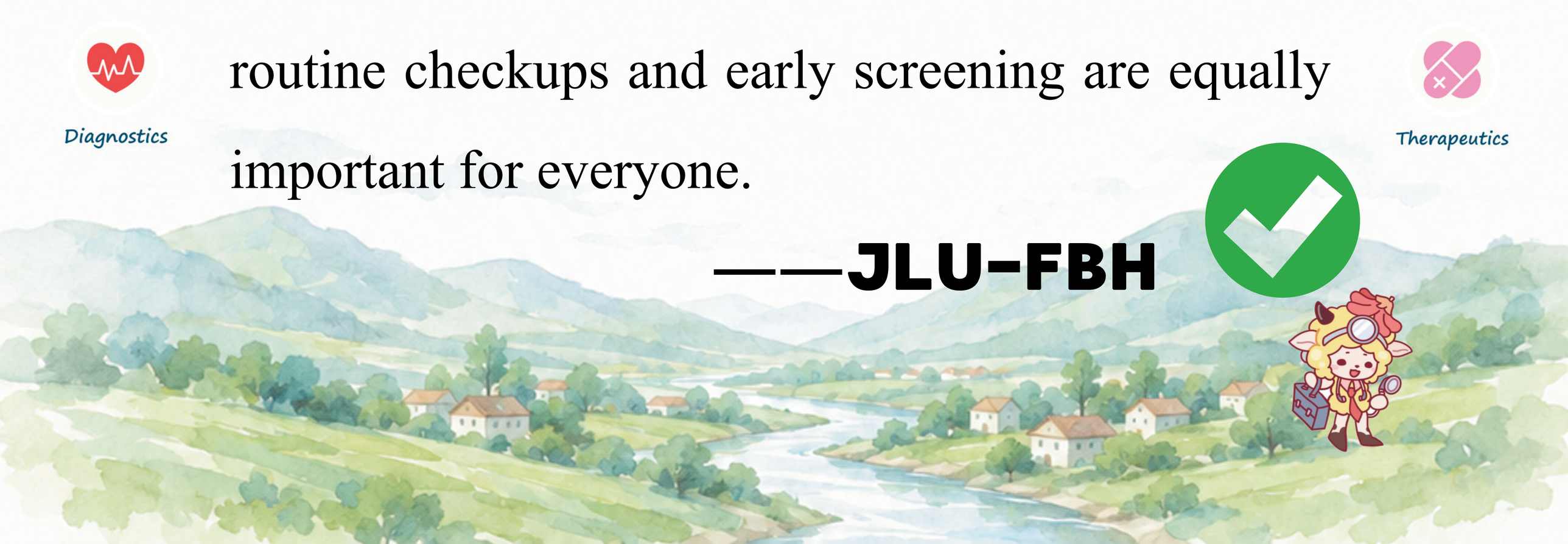
Therapeutics

Breast cancer is hereditary—if no one in the family has had it, I won't get it either.

The Truth:

Only about 5%~10% of breast cancers are associated with clear inherited genetic mutations. The vast majority, about 90% are sporadic, related to age, hormone levels, lifestyle, environmental factors, and other acquired factors. No family history does not mean zero risk—routine checkups and early screening are equally important for everyone.

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Gut microbiota has nothing to do with breast tumor development—it's mainly about genetics and hormones.

The Truth:

Recent studies have revealed that gut microbiota can influence breast tumor development and progression through the gut-mammary axis. Dysbiosis can lead to systemic chronic inflammation, abnormal estrogen metabolism, and impaired immune surveillance, thereby affecting the breast tumor microenvironment. Moreover, gut microbiota composition can influence the efficacy of immunotherapies such as immune checkpoint inhibitors. Maintaining gut microecological balance is therefore potentially significant for breast cancer prevention and treatment.

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As long as you maintain a healthy lifestyle, you won't get breast cancer—biological rhythms don't matter.

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The Truth:

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Chronic circadian disruption from long-term night shift work or sleep deprivation has been classified as a Group 2A carcinogen by the WHO. Circadian disruption interferes with normal melatonin secretion—which has antioxidant and anti-estrogenic effects—while abnormal expression of clock genes directly affects cell cycle regulation and DNA damage repair. Epidemiological studies show that women engaged in long-term night shift work have significantly elevated breast cancer risk. Circadian rhythm is a widely overlooked but important dimension in tumorigenesis.

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Wearing underwire bras causes breast cancer.

The Truth:

This is a classic myth repeatedly debunked but consistently resurging. No large-scale epidemiological studies or clinical evidence demonstrate a causal relationship between underwire bras and breast cancer. Major breast cancer risk factors include age, genetic mutations, hormone levels, reproductive history, alcohol consumption, and obesity. Underwire bras do not compress the lymphatic system or cause toxin accumulation—this claim has no basis in anatomy or physiology. Bra selection should prioritize comfort without panic-driven "risk avoidance."

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Breast nodules can be eliminated through massage and manipulation without the need for hospital examination or treatment.

The Truth:

Breast nodules are classified as benign or malignant. Before a pathological diagnosis is confirmed, blind massage may physically compress malignant nodules, increasing the risk of local spread or hematogenous/lymphatic metastasis. Improper massage may also cause microvascular rupture and hematoma formation within benign nodules, interfering with subsequent imaging evaluation. The correct approach upon discovering a breast nodule is to promptly seek medical attention, undergo breast ultrasound, mammography, or core needle biopsy to determine its nature, and then decide on the appropriate management strategy.

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Chemotherapy side effects like hair loss and nausea are just minor inconveniences during treatment—heart damage is not a concern.

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The Truth:

Conservation

Many commonly used breast cancer chemotherapeutic agents have well-established cardiotoxicity, causing irreversible myocardial damage and heart failure that may not manifest until years after treatment—known as "late-onset chemotherapy-induced cardiotoxicity." Additionally, radiotherapy to the chest area increases the risk of coronary artery disease. Therefore, long-term management of breast cancer patients must include not only tumor surveillance but also lifelong cardiovascular follow-up—an issue easily overlooked by the public and even some primary care providers.

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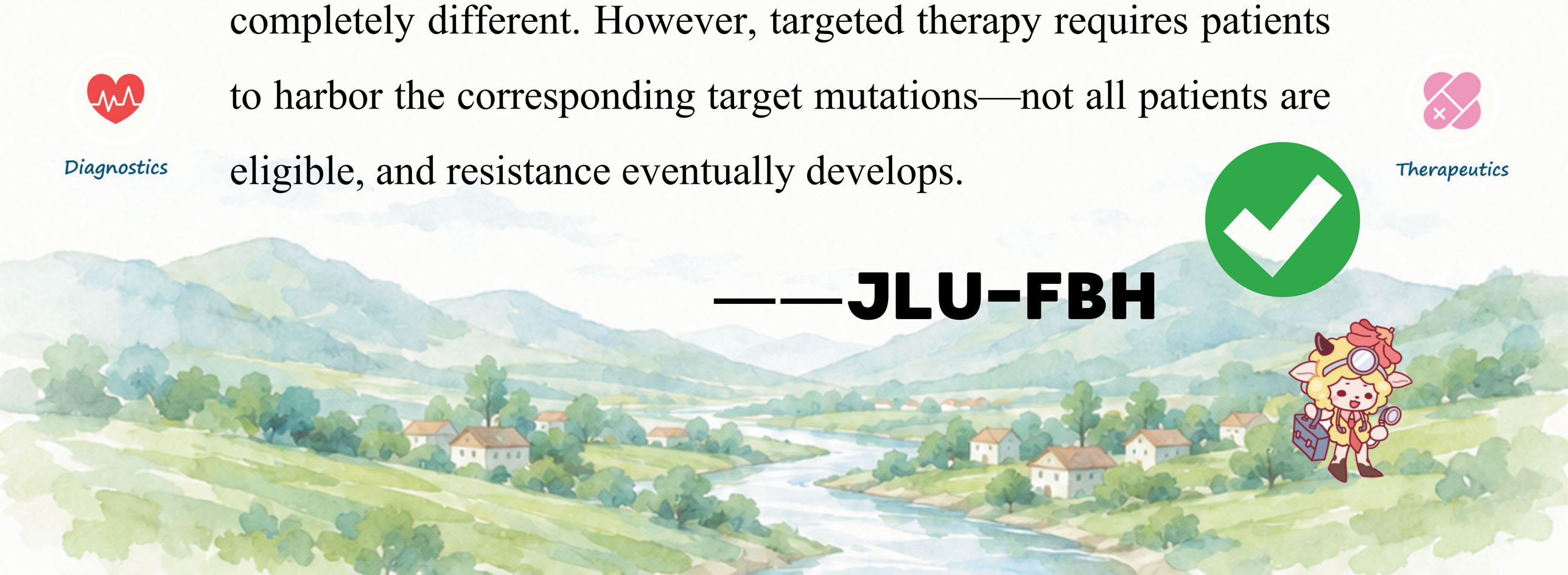
Therapeutics

Targeted therapy is just chemotherapy by another name—it kills both healthy and cancerous cells the same way.

The Truth:

Targeted therapy and chemotherapy are fundamentally different in mechanism. Chemotherapy kills all rapidly proliferating cells by interfering with DNA replication or cell division, lacking molecular selectivity. Targeted drugs are designed against tumor-specific driver mutations or molecular markers and theoretically only attack tumor cells carrying the specific target, with minimal impact on normal cells. Their side effect profiles are also completely different. However, targeted therapy requires patients to harbor the corresponding target mutations—not all patients are eligible, and resistance eventually develops.

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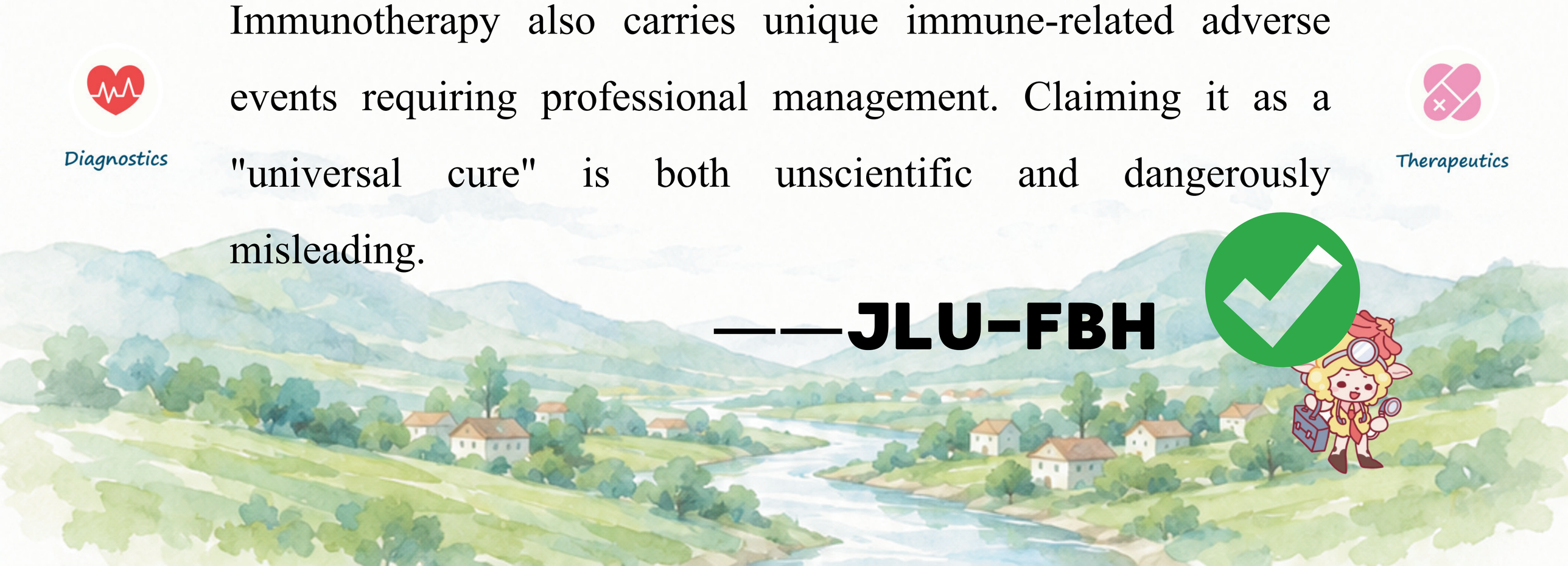
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PD-1/PD-L1 immunotherapy is a universal cure that works for all types of cancer.

The Truth:

PD-1/PD-L1 inhibitors represent a major breakthrough in tumor immunotherapy but are far from universal. Their efficacy highly depends on tumor microenvironment characteristics, particularly biomarkers like tumor mutational burden, PD-L1 expression, and microsatellite instability. Clinical data show objective response rates vary dramatically across cancer types and patient populations—and are nearly zero in some "cold tumors." Immunotherapy also carries unique immune-related adverse events requiring professional management. Claiming it as a "universal cure" is both unscientific and dangerously misleading.



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Radiation therapy "burns" cancer cells—patients become radioactive and cannot safely be around family members afterward.

The Truth:

Radiotherapy uses high-energy rays to locally irradiate tumors and induce DNA double-strand breaks to kill tumor cells. Modern precision techniques concentrate radiation dose highly on the tumor target while minimizing exposure to surrounding normal tissue. After treatment, radiation energy does not remain in the body—patients do not become sustained "radiation sources" and can safely have normal contact with family members, including pregnant women and children. This misconception leads many patients to reject necessary radiotherapy due to fear, causing preventable local recurrence.



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circRNAs have no functional difference between normal and tumor cells. They are simply "useless" non-coding RNAs.

The Truth:

circRNAs are a class of covalently closed non-coding RNAs formed by back-splicing and are widely present in normal cells. However, in tumor cells, many circRNAs are aberrantly expressed. Some circRNAs play dual regulatory roles (pro-tumor or anti-tumor) in breast cancer and other malignancies through mechanisms such as miRNA sponging, transcriptional regulation, or protein interactions.

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Vaccines only work against infectious diseases and are meaningless for cancer.

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The Truth:

Conservation

Cancer vaccines are an important branch of tumor immunotherapy, classified into preventive vaccines and therapeutic vaccines. Preventive vaccines such as HPV vaccine can prevent cervical cancer and other HPV-related cancers. Therapeutic vaccines target cancer patients by delivering tumor-specific antigens to activate the patient's own immune system to recognize and kill tumor cells.

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Cancer can spread from person to person through everyday contact, such as shaking hands or sharing meals.

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The Truth:

Cancer itself is generally not contagious and cannot spread through everyday contact, such as shaking hands, hugging, sharing meals, or breathing the same air. Although certain viruses and bacteria can increase the risk of developing specific cancers, it is the pathogen—not the cancer itself—that may be transmitted between people.

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Air2Protein

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Proteins produced from air are chemically synthesized substances with safety risks similar to industrial chemicals.

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The Truth:

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This approach is not conventional chemical synthesis but a hierarchical process coupling electrocatalysis and biocatalysis: electrical energy drives catalysts to reduce N_2 and CO_2 into small precursors (e.g., formate, ammonia), which are then assembled into amino acids and proteins via engineered enzymes or cell factories. The final proteins are identical in sequence and conformation to natural counterparts, not artificial polymers. The process runs under mild aqueous conditions, with intermediates and products monitored by chromatography–mass spectrometry, and final products undergo food-grade purification and toxicological evaluation. Hence, they are bio-manufactured food ingredients, with safety assessment comparable to that of fermentation-derived protein products.

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If the pollutant is no longer detectable, the pollution problem is considered completely solved.

The Truth:

A decrease in pollutant concentration does not equate to complete elimination. During the degradation of many organic pollutants (such as polycyclic aromatic hydrocarbons, PAHs), toxic intermediate products may be generated, leading to secondary pollution. Moreover, pollutants may simply transfer from one medium to another without being truly removed from the environment. Genuine remediation requires complete mineralization – the thorough conversion of organic compounds into CO_2 and water – rather than merely making the pollutant "disappear."

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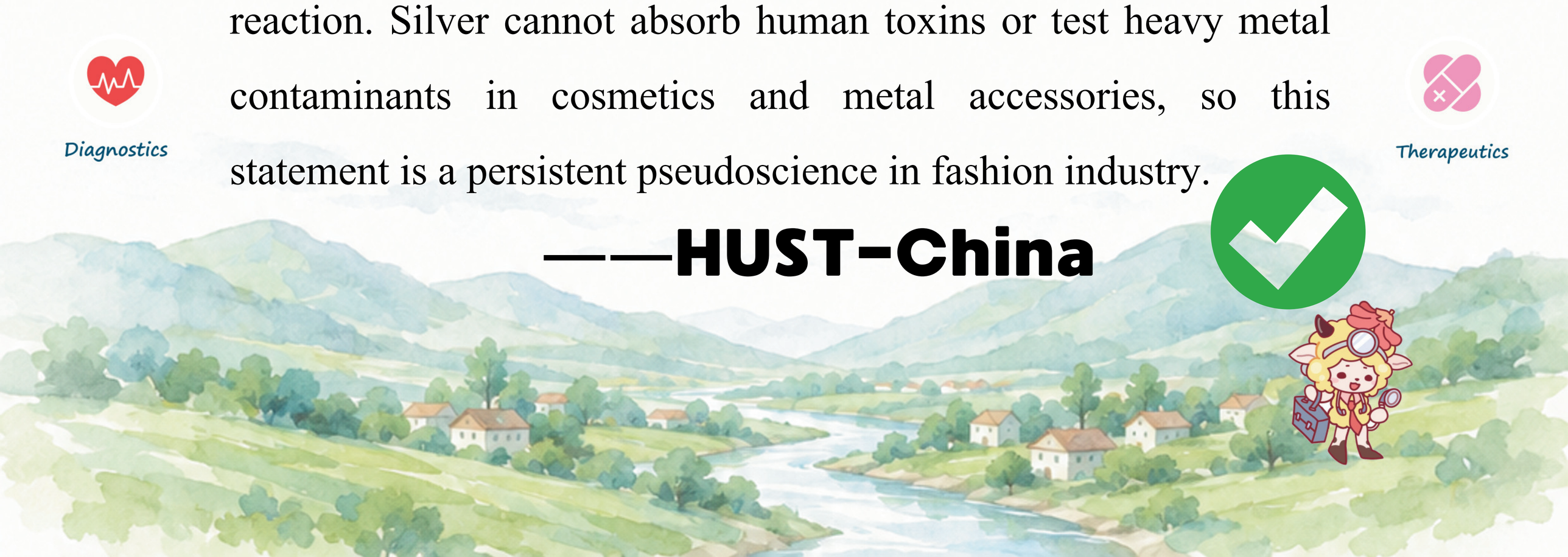
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Silver will turn black when contacting toxic substances; blackened silver jewelry can absorb toxins from human body and detect heavy metal poisons in cosmetics and ornaments.

The Truth:

The blackening of silver products results from the chemical reaction between silver and sulfur to form silver sulfide, which has no connection with toxicity. In ancient times, only crude arsenic trioxide mixed with sulfur impurities could darken silver, while purified arsenic trioxide causes no discoloration. Sulfur exists in sweat, perfume and fabric dyes, which can all oxidize silver. The tarnish of silver jewelry is merely a common chemical reaction. Silver cannot absorb human toxins or test heavy metal contaminants in cosmetics and metal accessories, so this statement is a persistent pseudoscience in fashion industry.

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Drinking carbonated beverages causes osteoporosis.

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The Truth:

Conservation

The concern that carbonated beverages cause osteoporosis arises from the phosphoric acid contained in drinks, which may slightly hinder calcium absorption. However, The established risk factors for osteoporosis include estrogen decline, smoking, and physical inactivity, rather than carbonated beverage consumption alone. The moderate consumption of carbonated beverages alongside a balanced diet does not cause osteoporosis.

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Influenza A is basically the same as a common cold, and it will pass without much concern.

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The Truth:

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Influenza A is an acute respiratory infectious disease caused by influenza A virus. It usually has a more abrupt onset, stronger systemic symptoms, higher transmissibility, and can lead to severe complications, especially in high-risk populations.

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Influenza spreads only when an infected person coughs or sneezes.

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The Truth:

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Influenza can spread through droplets, contact, and potentially aerosols. In crowded indoor spaces such as subways, hospitals, and classrooms, infection risk may exist even without obvious coughing.

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Antigen tests are always enough, and PCR testing is unnecessary.

The Truth:

Antigen tests are useful for rapid screening, but nucleic acid testing is more sensitive and specific, especially in early infection, low viral load cases, and pediatric diagnosis.

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Children with flu do not necessarily need testing; observation is enough.

The Truth:

Because children may deteriorate more quickly and symptoms can overlap with other respiratory infections, nucleic acid testing helps confirm diagnosis early and supports proper treatment and isolation.

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Only hospital testing matters; environmental monitoring is unnecessary.

The Truth:

Clinical testing confirms infection after exposure, while environmental surveillance can provide early warning before outbreaks spread. The two approaches are complementary.

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Any genetically engineered yeast used in public spaces is inherently dangerous.

The Truth:

Engineered yeast is not a pathogen or “artificial virus.” With proper containment, biosafety review, and inactivation measures, it can be used safely in enclosed devices for sensing applications.

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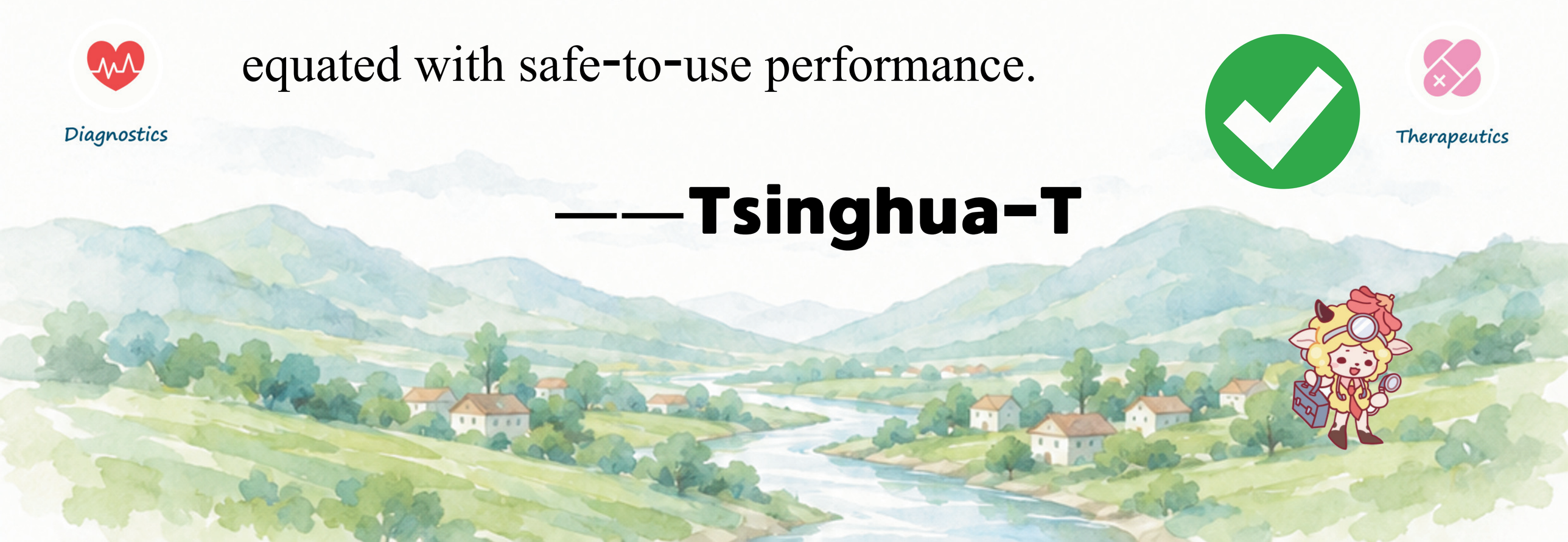
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Skincare ingredients produced by bio-manufacturing are inherently safe and risk-free.

The Truth:

Skincare ingredients derived from bio-manufacturing do not equal absolute safety. Even microbially-biosynthesized active ingredients need comprehensive safety, sensitization and toxicity assessments before being applied to skincare products. “Biological origin” cannot be directly equated with safe-to-use performance.

——Tsinghua-T



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Synthetic multicellular systems do not fundamentally differ from simple mixtures of multiple cell populations.

The Truth:

Compared with directly mixing multiple cell types, artificial cell differentiation systems allow a single founder cell population to undergo programmable, dynamic, and stable cell fate branching rather than depending on externally assembled cell combinations.

——WHU-China

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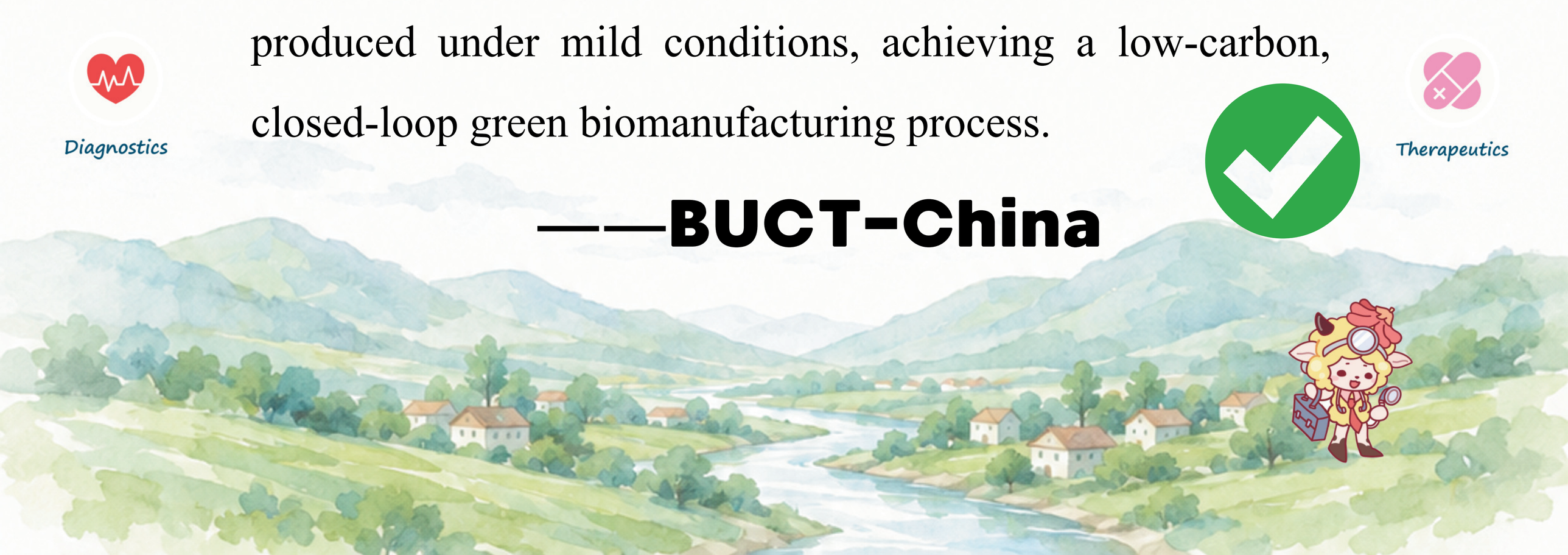
Therapeutics

The manufacturing of transparent materials must rely on petrochemical feedstocks and extreme high-temperature/high-pressure conditions, and bio-based raw materials cannot replace traditional processes to produce materials with high transparency.

The Truth:

We propose a purely biological approach: CGTase enzyme converts starch from food waste into β-cyclodextrin (host molecule), while engineered E. coli converts glucose into L-homoserine (guest molecule) through rewired metabolic pathways. Through host-guest complexation and hot pressing, transparent materials with high optical clarity, mechanical robustness, and reprocessability can be produced under mild conditions, achieving a low-carbon, closed-loop green biomanufacturing process.

——BUCT-China



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RNA-based pesticides (RNAi pesticides) will alter human genes and cause harm to the human body.

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The Truth:

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RNAi will not alter human genes and is safer than most conventional pesticides. RNAi (RNA interference) refers to a natural gene-regulatory mechanism.The mechanism of RNAi is not to modify genes, but to regulate gene expression. RNA sequences transcribed by different organisms are distinct. Essentially, the active ingredient of RNAi pesticides is RNA. Most RNA residues on the surface of food can be removed by washing with water.

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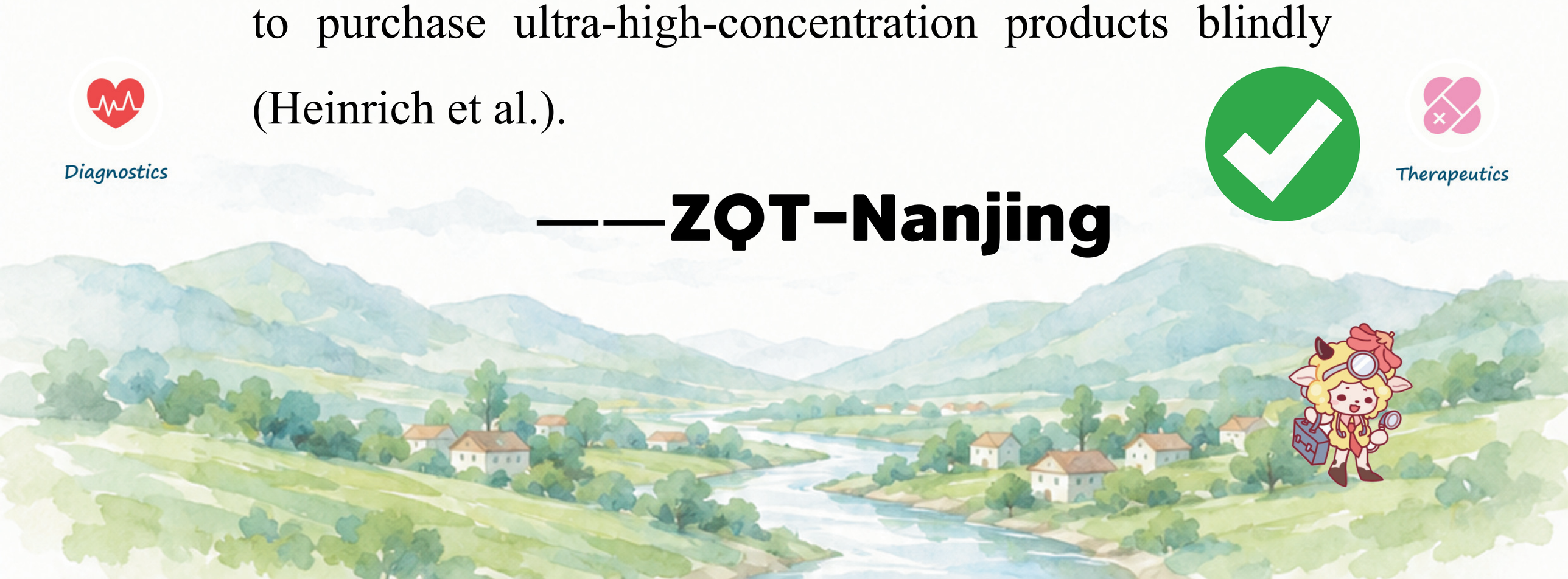
Therapeutics

Higher concentrations of ectoin bring stronger moisturizing effects.

The Truth:

A concentration of 0.5%–1% fully meets the demands of daily skin moisturization and seasonal skin stabilization. For repairing skin after acid exfoliation or wound care following cosmetic procedures such as microneedling and photorejuvenation, a maximum concentration of 1%–2% is sufficient. There is no need to purchase ultra-high-concentration products blindly (Heinrich et al.).

——ZQT-Nanjing



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Only membrane-bound organelles can form independent functional compartments within the cell.

The Truth:

Membraneless compartments also exist within cells. Proteins and nucleic acids can form condensates through liquid-liquid phase separation, enriching specific molecules in localized regions and thereby creating microenvironments with distinct functions. Thus, spatial compartmentalization in cells does not necessarily require membrane structures.

——NAU-CHINA



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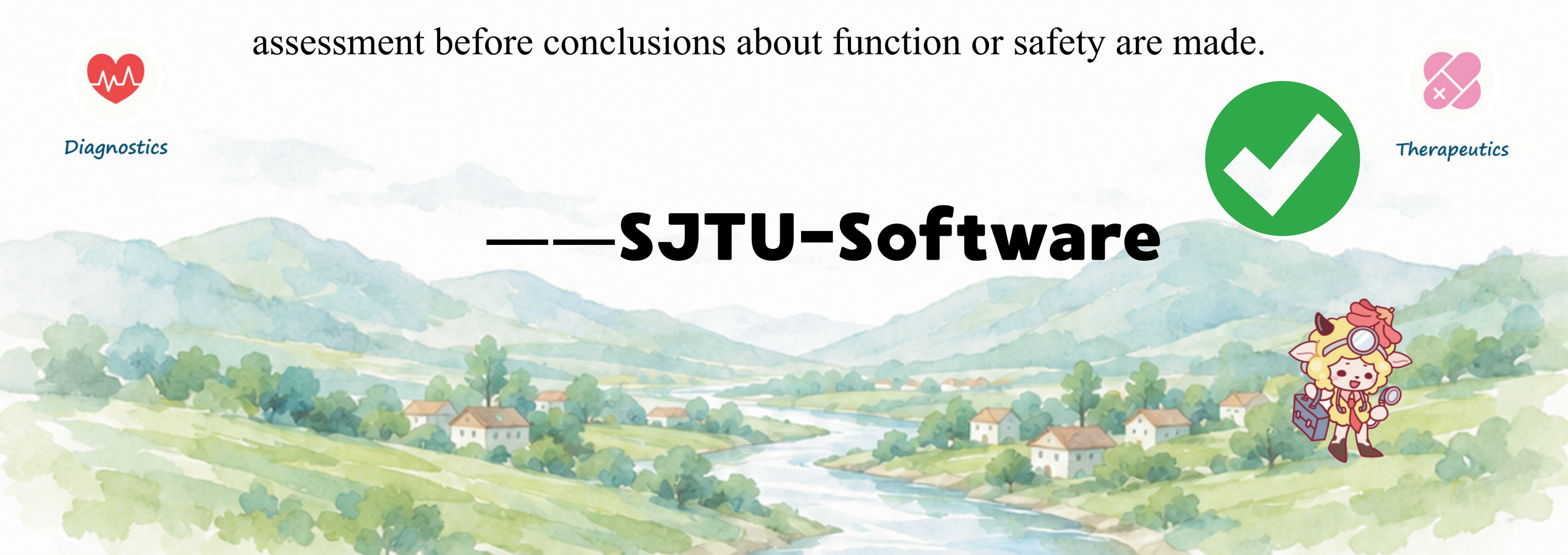
Therapeutics

High accuracy in computational tests, or a high score assigned to a biological design by an AI model, guarantees that the proteins, metabolic pathways, or genetic circuits it designs will function safely in living systems; wet-lab validation is therefore unnecessary.

The Truth:

AI outputs are computational predictions shaped by training data, optimization objectives, and model assumptions, not experimental facts. Even strong performance on an internal test or benchmark may not generalize to out-of-distribution data or complex cellular contexts. AI-designed biological systems therefore require representative independent testing, appropriate benchmarking, context-relevant wet-lab validation, and biosafety and risk assessment before conclusions about function or safety are made.

——SJTU-Software



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AI can now design functional viruses from scratch, so anyone can easily use this technology to create bioweapons.

The Truth:

Researchers have used AI to design bacteriophages—viruses that infect bacteria—but this does not mean that anyone can easily create dangerous viruses that infect humans. Generating a genome sequence is not the same as obtaining a functional virus: specialized laboratory work, equipment, and experimental validation are still required. This research demonstrates the potential of AI-assisted genome design, while risk assessment, DNA synthesis screening, and safeguards against misuse remain important.

——DKU



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Tree deaths are all caused by drought or insect infestation.

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The Truth:

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Drought and insect damage are only two of many possible causes. Pathogenic microorganisms, such as fungi, bacteria, and nematodes—can also directly kill trees. Tree mortality is the result of multiple interacting biotic and abiotic factors.

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NACIS SHANGHAI

Since the proteins inside biological fertilizers are “poisonous” towards some pests, does this mean that it will hurt human organs also?

The Truth:

No, take the common BT protein pesticide as an example, the protein is activated in the alkaline intestines of the pests and binds to specific receptors, destroying cell membranes and causing fatal damage towards specific insects. For mammals, including human, such protein does not have severe effects on the organs or cells, which means it would be completely safe for people to ingest. (except for consuming tons of it, which may cause diarrhea or indigestion.)

——NACIS SHANGHAI

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If BT protein will effect the pests, does this mean that other insects may also be affected by such pesticides?

The Truth:

No, in fact, only a certain type of pesticide is able to affect a certain type of pest. For instance, the most effective type of BT protein that aims for Lepidoptera insects is Cry-1A, which means this protein may not still be functional on other insects with other orders. This suggests that if using the toxic protein that is only useful for one certain type of pest, the other insects (possibly beneficial insects) will not be disturbed and still wholesome in their habitat.

——NACIS SHANGHAI

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If biological pesticides are also considered as “pesticides”, does this mean that the problem with pesticide resistance still occurred by biological pesticides, similar to chemical pesticides?

The Truth:

No， take BT protein as an example, it is combined with multiple receptors on the cell membranes of the offsprings of the, which means that it is not easy for the insect to adapt under such conditions. There are also fungi pesticides and motivative pesticides, which either is having inevitable functions (kill pests by permeability and parasitism) or helps the self-defensive system of plants themselves, which means the possibility of adapting such resistance is very low.

——NACIS SHANGHAI